

Jack Leckert

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PROFILE

Software Engineer at Swift Navigation and UC Berkeley graduate, specializing in building and deploying production-grade software for autonomous systems, localization, perception and controls.

RESEARCH INTERESTS	TECHNICAL SKILLS	PROFESSIONAL SKILLS
Machine learning Model and learning based controls Localization	Python, C/C++, Go, TypeScript Linux / CLI Github / CI AWS, Kubernetes, Jenkins	French – native language English – fluent German – fluent

PROFESSIONAL EXPERIENCE

Software engineer <i>Swift Navigation, San Francisco, CA</i> Software engineer as part of the IoT team. - Proposed, led and built an automated infrastructure monitoring platform to optimize the quality and availability of Swift's base stations network. This effort led to the acquisition of 2 new customers. - Fine-tuned Starling positioning engine parameters using a genetic algorithm, leading to a 20% increase in 2D position accuracy of Swift's XGPS DashPro. - Designed and built a suite of customer-facing REST APIs for real-time service indicators (geofencing, coverage). - Developed various protocol converters for data processing and visualization.	JULY 2024 – PRESENT
Graduate student instructor <i>UC Berkeley, CA, USA</i> Teaching assistant for the undergraduate class Experimentation and Measurements. Preparing and supervising weekly labs of 20+ students, along 2 other GSIs.	AUGUST 2023 – JUNE 2024
Flight software engineer <i>Skybase, Christchurch, New Zealand</i> Flight software engineer with the mission to develop a vision system for an autonomous fire retardant airplane. Developed automated object detection functions for a remotely piloted flight control system.	FEBRUARY 2023 – JUNE 2023
Technology analyst intern <i>Starburst Aerospace, Munich, Germany</i> Technology analyst intern as part of the Starburst Startup Accelerator in Munich. - Conducted one major consulting project about hot and harsh environment electronics for a European Tier-2 Aerospace supplier. - Assessed the technology viability of Aerospace startups and fit for the Starburst community (+300 startups).	MAY 2022 – OCTOBER 2022

EDUCATION

Master of Engineering, Robotics <i>University of California, Berkeley, USA</i> Focus on controls and machine learning. GPA: 3.96/4.00.	AUGUST 2023 – JUNE 2024
Engineering degree, Robotics & Mechanics <i>École Centrale de Lyon, France</i> Computer science, machine learning, non-linear automatic control, numerical analysis, mathematics, solid and fluid mechanics, physics, material science, signal processing, electronics. GPA: 3.78/4.00.	SEPTEMBER 2020 – JUNE 2022

Preparatory classes for the Grandes Ecoles

SEPTEMBER 2018 – JUNE 2020

Lycée du Parc, Lyon, France

Two years of intensive preparatory classes for the French engineering schools nationwide scientific competitive examination (MPSI and PSI*).

Mathematics, physics, mechanics, automatic control, computer science, chemistry.

High School

SEPTEMBER 2015 – JULY 2018

Lycée Français Victor Hugo, Frankfurt, Germany

Scientific Baccalauréat (high school diploma), highest honours.

One trimester of the 10th grade spent at Winnipeg, Canada.

RESEARCH AND ENGINEERING EXPERIENCE

UAV research project

SEPTEMBER 2023 – JUNE 2024

UC Berkeley, Hybrid Robotics lab

Research project focusing on developing a controller for a flapping wing UAV using model-based control and reinforcement learning. Simulation software used is Mujoco.

- Drone design developed on Solidworks and converted to XML.
- Computation of fluid forces in C++ using aero strip theory (Wagner's model).
- MPC and RL controller developed in Python for the simulation.

ESA AutoICE challenge

FEBRUARY 2023 – MAY 2023

Deep learning challenge in partnership with DTU to automate sea ice mapping of the Arctic using Sentinel-1 SAR imageries.

- Selection of three main parameters to qualify: ice concentration, stage of development and floe size.
- Trained a U-net (contracting and expanding CNN) using parallel cloud computing from Azure.
- By fine-tuning hyperparameters, I reached F1 and R2 scores of about 80%.

Fluid mechanics research project

SEPTEMBER 2021 – MAY 2022

École Centrale de Lyon, Fluid Mechanics and Acoustics Laboratory (LMFA)

Research study focusing on reducing the drag force on a marathon runner by placing partners in a specific formation around him.

- Designed and machined a test bench using CAD and Fusion 360.
- Tested experimentally and analyzed the effects on the drag force when varying parameters (distance and angles) between the runners.
- Designed a formation by iteration allowing to save up to 2min 23s on a full marathon, or 27s less than the actual world record from E. Kipchoge during the INEOS 1:59 Challenge in 2019.

A research paper has been published in the Proceedings of the Royal Society A.

Aerospace engineering project

SEPTEMBER 2020 – JUNE 2021

École Centrale de Lyon, Fluid Mechanics and Acoustics Laboratory (LMFA)

Building of a supersonic rocket (the Centrale Transonic Rocket) with a team of 5 students as part of the national competition C'Space organised by Planète Sciences.

- Designed and built an electronic acquisition chain to save pressure data using C++.
 - CAD designed and 3D-printed a rocket cone.
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VOLUNTEERING AND ASSOCIATIVE EXPERIENCE

President of a student organization

OCTOBER 2020 – FEBRUARY 2022

Centrale Lyon Conseil, Lyon, France

Centrale Lyon Conseil is the Junior-Enterprise of École Centrale de Lyon, the analogue of a consulting club. Students apply their theoretical and practical skills to carry out projects in various engineering fields and work for a wide range of companies: startups, small or big enterprises and communities.

- Managed a team of 24 students in order to reach the goals of the strategy.
- Managed around 15 customer projects, including websites, engineering projects and market studies, resulting in a 150% increase in sales.